

Incidence of clinically significant POPF depending on the type of pancreatic remnant and anastomosis type.

Procedure	#	POPF(%)	P value
PJ, low risk	66	3(4.5%)	0.005*
PJ, high risk	25	8(32%)	
PG, high risk	48	4(8.3%)	0.016**

*PJ, low risk vs. PJ, high risk; **PJ, high risk vs. PG (no PG done for low risk)

Mo1341

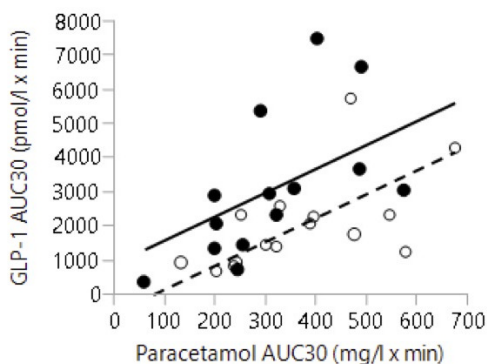
GASTRIC EMPTYING AND MODE OF OPERATION INDEPENDENTLY ENHANCE THE RELEASE OF GLP-1 AFTER A MIXED MEAL AND IMPROVE GLYCEMIC CONTROL IN SUBJECTS HAVING UNDERGONE WHIPPLE'S PROCEDURE AND PYLORUS PRESERVING PANCREATODUODENECTOMY (PPPD)

Johannes Miholic, Emanuel Steiner, Lukas Kazianka, Marlene Wewalka, Birgit Reiter, Jens Holst

Introduction. Gastric emptying and postprandial release of the incretin GLP-1 are enhanced after pancreaticoduodenectomy (PD), which is an operation for tumors of the pancreas head or papilla. New onset diabetes mellitus is frequent in pancreas cancer and resolves to a great deal after PD, the mechanism of which is not well understood. GLP-1 conceivably plays a role. Rapid gastric emptying and malabsorption might trigger GLP-1 release which is induced by the arrival of undigested food in the distal gut. It was the purpose of this study to shed light on factors associated with GLP-1 release after a mixed meal in subjects having undergone PD. **Patients and methods.** 15 subjects who have undergone pylorus preserving pancreaticoduodenectomy (PPPD) and 15 after PD with distal gastrectomy (Whipple procedure) were studied 3-118 (median: 12) months after the operation. Median age was 62 years, the male female ratio 1, and Body mass index 22.7. All subjects ingested a liquid test meal containing 1 g paracetamol solved in water, taking the area under the paracetamol plasma concentration curve in the first half hour as a measure of gastric emptying. Before the meal and 10, 20, 30 minutes thereafter venous blood was drawn for the measurement of paracetamol, glucose, insulin, and GLP-1. Variables are shown as median and range, tested for significance using Wilcoxon's nonparametric test for continuous variables. Univariate and multiple linear regression were carried out where appropriate. **Results.** Fasting glucose, HbA1c, and the early (0-30 min.) integrated concentrations of paracetamol and GLP-1 are listed in Table 1. In univariate analysis gastric emptying correlated closely with early GLP-1, whereas the difference of GLP-1 in the two modes operations was not significant. Multiple regression, however revealed (p=0.0015) gastric emptying (p=0.015) and Whipple vs. PPPD (p=0.0012) to be independent significant predictors of early GLP-1 release (Figure1). The antidiabetic effect of GLP-1 is illustrated by the significantly lower HbA1c measurements in Whipple subjects (Table 1). **Conclusion.** In pancreaticoduodenectomy, resection of the distal stomach results in more rapid gastric emptying and accelerated small intestinal nutrient exposure which probably explain the enhanced postprandial release of GLP-1 and the superior glycemic control after the Whipple procedure.

Table: Fasting Glucose and HbA1c, and gastric emptying (paracetamol AUC30) and GLP-1 AUC 30 in Whipple and PPPD subjects.

	Whipple procedure	Pylorus preserving PD	P-value
Fasting glucose (mg/dL)	90(82-141)	100(86-151)	0.13
HbA1c	5.3(4.6-6.2)	5.8(4.8-7.1)	0.009
Paracetamol AUC30	300(61-573)	330(133-674)	0.38
GLP-1 AUC30	2908(360-7460)	1750(675-5710)	0.11



Multiple regression diagram of gastric emptying and GLP-1 release in PPPD (dotted line) and Whipple (solid line) subjects.

Mo1342

TARGETABLE DISPARITIES IN THE SURGICAL TREATMENT OF EARLY-STAGE PANCREATIC CANCER IN KENTUCKY

Heather A. Frohman, Quan Chen, Bin Huang, Anh-Thu Le, Sean Dineen, Jeremiah T. Martin, Ching-Wei Tzeng

Introduction: There has been a national failure to operate on more than 50% of patients with resectable pancreatic adenocarcinoma (PDAC). The primary aims of this study were to evaluate surgery rates and identify targetable disparities in resectable PDAC patients in the state of Kentucky. **Patients and Methods:** Patients with clinical stage I-II PDAC in the Kentucky Cancer Registry (2004-2013) were included. Multivariate modeling was used to identify factors associated with surgery received and overall survival (OS). An academic hospital (AH) was defined as one of the four main campus hospitals staffed by faculty from either of the two allopathic medical schools in Kentucky. **Results:** Of 1,680 in-state patients diagnosed with stage I-II PDAC over the study period, only 849 (50.5%) underwent curative-intent resection. Resection rates did not change over the decade studied (range 44.5-55.7%, p=0.58). Only 45.6% of patients had treatment at an AH. There were no time trends in referral rates to AH. AH were more likely to treat patients with the following traits: Black, Appalachian county, active smoker, uninsured or Medicaid, counties with lower education levels and greater poverty (all p<0.039). Total resections were almost evenly split between AH and non-AH with 53.4% occurring at an AH. However, the resection rate was significantly higher at AH vs. non-AH (59.1% vs. 43.3%, p<0.001). Variables independently associated with resection were AH (OR 1.64, p<0.001), age (20-49, 50-64, 65-74, vs. 75+ years, OR 5.54, 3.15, 2.54, p<0.001 for each), stage I vs. II (OR 2.11, p<0.001), and residing in less impoverished counties (low vs. very high poverty, OR 1.66, p=0.01). Gender, insurance type, county education level, and Appalachian county were not significantly associated with resection rate. Pre-treatment clinical factors independently associated with improved OS included resection (hazard ratio, HR 0.35, p<0.001), younger age (age 65-74 vs. age 75+, HR 0.77, p<0.001), and early stage (stage I vs. II, HR 0.71, p<0.001). **Conclusions:** The most significant disparity influencing resection rates in early stage PDAC in Kentucky is facility type, with AH being 64% more likely to resect patients, even when accounting for differences in patient demographics. Surgical resection remains the most important determinant of OS in stage I-II PDAC. In Kentucky, coordinated statewide efforts to improve regionalization of care to academic hospitals may improve both resection rates and OS for patients with early-stage PDAC.

Mo1343

TRANSDUODENAL RESECTION FOR DUODENAL ADENOMAS MAY BE UNDER UTILIZED TOOL

Edward E. Cho, Peter Deleeuw, Alyssa J. Chapates, Houssam Osman, D Rohan Jeyarajah

Introduction Duodenal adenomas (DAs) are uncommon precancerous pathologies with malignant potential. Currently transduodenal resection and the Whipple procedure remain the surgical options for these lesions. The optimal surgical management of DAs remain controversial. **Method** We performed a retrospective review of 27 patients at our institution who received either a Whipple procedure (40.7%) or a transduodenal resection (TDR) (59.3%) for ampullary and nonampullary duodenal adenomas. Perioperative and postoperative outcomes were compared between the two groups. **Results** Both groups had similar demographics. Median operative time and median estimated blood loss (EBL) were longer in the Whipples group than the TDR group. Mean length of stay was longer in the Whipple group vs TDR group (12 vs 7 days). 18% of patients who received the Whipple procedure subsequently developed a subclinical pancreatic leak controlled with a drain. No bile leak was seen from the duodenotomy with the TDR patients. 3 patients (18.7%) who received TDRs developed recurrent adenomas. Two of these patients subsequently received a Whipple procedure. This is summarized in Table 1. **Conclusion** TDR has the perioperative advantages of less operative time, less EBL less hospital length of stay and a low postoperative morbidity. Although recurrences are possible after TDR, majority of our patients who underwent TDR are adenoma-free without being subjected to the morbidity and mortality of the Whipple procedure. Due to the safety and potentially curative benefit, TDRs should be implemented as part of the clinical algorithm in treating selective patients with duodenal adenomas.

Table 1

	Whipple (n=11)	Transduodenal Resection (n=16)
Median Duration of Surgery (min)	248	120
Median Estimated Blood Loss (mL)	300	100
Mean Length of Stay (Days)	13	7
Leak (%)	18%	0%
Recurrences	N/A	18.7%

Mo1344

PANCREATIC FISTULA OR LEAK AFTER DISTAL PANCREATECTOMY: HAS ANYTHING CHANGED? 13-YEAR SINGLE SURGEON EXPERIENCE

Anthony Morada, Pratik Mehta, Braden N. Miller, Alagappan A. Annamalai, Ashley Wachsmann, Srinivas Gaddam, Laith H. Jamil, Simon K. Lo, Nicholas N. Nissen

Background: Pancreatic fistula (PF) after distal pancreatectomy (DP) continues to pose a substantial clinical problem. **Objective:** To evaluate a single-surgeon experience with PF following DP. **Design:** Academic medical center. Review of surgical database. PF diagnosed by strict ISGPF criteria. Consecutive patient cohort evaluated in total and also as early (n=68) and late (n=68) cohorts. **Results:** 136 patients underwent DP between 2003 and 2016. Pancreatic stump was managed by stapled (n=109) or pledgetted closure (n=23). Peri-pancreatic drains were used routinely. Patients in the more recent cohort were older (65 vs 60 years, P= .04), and were more likely to have adenocarcinoma (p=.04) and to undergo laparoscopic resection (p<.01). Overall morbidity and mortality were 27% and 1%. PF occurred in 44 patients (32%) including 21 grade A, 22 grade B, 1 grade C. In 21 patients with PF, surgical drainage provided definitive management with drain removal on mean

post-op day 23. In the remaining 23 patients, either percutaneous drainage (n=11) or endoscopic transampullary stenting/cystgastrostomy (n=12) was utilized as salvage therapy. No particular risk factor for PF could be identified in multivariate analysis except concurrent colon resection (3 of 6 patients). PF rates did not differ between early and late groups, but the use of endoscopic therapy was more common in the later group (p=.03). Those undergoing endoscopic salvage therapies also had a trend towards requiring more overall interventions (mean 3.7 vs 2.2 in those managed with percutaneous therapy, p=.03). Conclusion: Despite changes in the patient mix and surgical approach, the incidence of PF after DP has remained relatively constant. Endoscopic management of PF is increasing, although this may require more overall procedures. This approach should continue to be scrutinized.

Mo1345

EFFECT OF NEGATIVE PRESSURE WOUND THERAPY ON WOUND COMPLICATIONS FOLLOWING PANCREATECTOMY

Michael Kunczewitch, Aaron Blackham, Clancy J. Clark, Rebecca M. Dodson, Gregory Russel, Edward Levine, Perry Shen

Introduction: Wound complications in the acute and chronic setting are a source of considerable morbidity and cost to patients and health care systems. The prophylactic application of negative pressure wound therapy (NPWT) at the time of primary skin closure has been used with increasing frequency in attempts to decrease these events. We investigated the effects of NPWT on short- and long-term wound outcomes in patients undergoing pancreatotomy. **Methods:** A randomized controlled trial comparing the effect of NPWT to standard surgical dressing (SSD) on wound outcomes was performed in 265 high-risk surgical oncology patients undergoing laparotomy from 2012 to 2016. There were 73 patients in the study who underwent pancreatotomy and were randomized to SSD or NPWT. Postoperative wound complications in the first 30 days and incisional hernia rates were assessed. Secondary outcomes evaluated were rates of pancreatic fistula and delayed gastric emptying. **Results:** In the SSD (n=37) and NPWT (n=36) cohorts, the superficial SSI, deep SSI, seroma, and dehiscence rates were 16% and 14% (p>0.99), 5% and 8% (p=0.67), 16% and 11% (p=0.74), and 5% and 3% (p>0.99), respectively. The rates of incisional hernia were 32% and 14%, between the SSD and NPWT groups, respectively (p=0.067). Pancreatic fistula (22% and 31%, p=0.43) and delayed gastric emptying (11% and 8%, p>0.99) also demonstrated no significant difference in the SSD and NPWT arms. These findings were true irrespective of the type of pancreatic resection performed. **Conclusion:** Both short- and long-term wound complications were not improved with NPWT after pancreatotomy. There was a trend towards decreased incisional hernia rates in patients treated with NPWT. Further study is needed to better assess the impact of NPWT on wound outcomes after pancreatotomy.

Mo1346

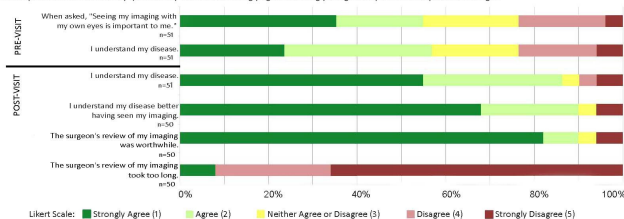
SURGEON-LED IMAGING REVIEW FOR PATIENTS WITH PANCREATIC AND RELATED MASSES AND A NEW QUALITY METRIC?

Sami Tannouri, Jessica A. Latona, Harish Lavu, Theresa P. Yeo, Jordan M. Winter, Charles J. Yeo

Purpose: An abundance of radiologic studies often accompanies patients at a surgical consultation. The perceived benefit of utilizing imaging as an educational tool for patients to facilitate a greater understanding of their complex medical diagnoses and enable them to actively participate in making decisions has not yet been studied. **Methods:** In this IRB approved study, we surveyed patients presenting for pancreaticobiliary surgical consultations at a tertiary care institution. Surveys were administered pre- and post-appointment, where a surgeon scrolled through CT and/or MRI scans with the patient and family. All scans were reviewed by our pancreaticobiliary team prior to the patient encounter. Using a 1 (strongly agree) to 5 (strongly disagree) Likert scale, patients reported their perceived importance of viewing the imaging studies and understanding of their medical condition. **Results:** Of fifty-one patients surveyed, 78% reported they had not previously seen their imaging and only 55% of patients thought that it was important for them to do so. On average, the surgeons spent 2.7 ± 1.9 minutes reviewing imaging studies with the patient and family. Selected survey question responses are depicted in Figure 1. On the post-visit survey, 90% and 86% of patients, respectively, responded positively to better understanding their disease and their planned operation having seen the imaging studies. Mean scores for the statements "I understand my disease" and "I understand my planned operation" significantly improved from 2.5 ± 1.2 to 1.7 ± 1.1 (p = 0.001), and 3.2 ± 1.2 to 1.8 ± 0.9 (p = 0.0004) after the consultation. Ninety percent of patients felt the imaging review was worthwhile, including 100% of patients that were ultimately not deemed appropriate surgical candidates. **Conclusions:** After the appointment, almost all patients felt that the imaging review with their surgeon was valuable and enhanced their understanding, although many did not have this opinion prior to the visit. Surgeons should routinely incorporate a short imaging review into patient encounters as it can be utilized as a powerful educational tool. Furthermore, in the current climate of focus on patient satisfaction, we propose that this practice be further analyzed as a possible quality metric.

Figure 1: Survey Responses

Patient responses to selected survey questions by Likert scale. Strongly agree to strongly disagree responses are depicted left to right.



Mo1348

POSTPRANDIAL CHOLECYSTOKININ CONCENTRATIONS AND [¹¹C]METHIONINE UPTAKE OF THE PANCREAS IN PATIENTS AFTER PANCREATICOUDENECTOMY

Emanuel Steiner, Lukas Kazianka, Robert Breuer, Wolfgang Wadsak, Jens Rehfeld, Johannes Miholic, Georgios Karanikas

Purpose: The uptake of [¹¹C]methionine ([¹¹C]MET) in the pancreas has been discussed in the context of exocrine and endocrine pancreatic function. Cholecystokinin (CCK) is a strong stimulator of pancreatic enzyme release. It was the aim of this study to examine possible relationships of CCK release with the uptake of [¹¹C]MET in the gland and with the velocity of gastric emptying. **Methods:** Nineteen tumor-free survivors after PD (age mean $\pm$ SD: 61 ± 8.7 yr; ten male, nine female) were given a mixed test meal. One gram paracetamol was ingested with the meal to evaluate the speed of gastric emptying. Paracetamol and CCK were measured before as well as 10, 20 and 30 minutes after ingestion. Simultaneously 800 MBq of [¹¹C]MET were administered and the activity (maximum tissue standardized uptake values [SUVmax]) over the pancreas was measured using PET-CT at 15, 30 and 60 minutes after injection. Areas under the curve (AUC) were calculated for SUVmax over 60 minutes (total integrated SUVmax[AUC₆₀]), as well as for paracetamol and CCK over the first 30 minutes after test meal consumption (AUC₃₀). **Results:** Gastric emptying velocity (paracetamol AUC₃₀, median and range: $217.7 [60.5 - 587] \text{ mg/l} \times \text{min}$) showed no correlation with postprandial CCK concentrations (AUC₃₀, median and range: $201 [77.5 - 552] \text{ pmol/l} \times \text{min}$). CCK AUC₃₀ correlated with total integrated SUVmax (AUC₆₀; $R^2=0.26$, $p=0.0307$, Figure 1). **Conclusion:** Since CCK is a well-known stimulant of the exocrine pancreas the association between postprandial CCK concentrations and pancreatic [¹¹C]MET uptake might suggest a causal relationship. To some extent [¹¹C]MET uptake could be used to assess the activity of the exocrine pancreas. Further research in that direction could provide a barely invasive method to assess the activity of the exocrine pancreas.

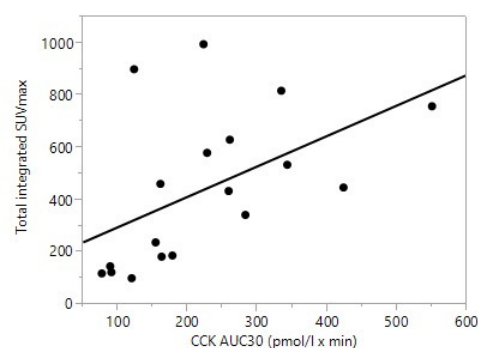


Figure 1: Relationship between total integrated SUVmax (AUC₆₀) and cholecystokinin AUC₃₀ in patients after PD

Mo1349

AN EXPLORATION OF REGIONAL AND RURAL-URBAN DIFFERENCES IN PANCREAS CANCER SURGERY

Erin MacKinney, Whitney Zahnd, Emily C. Sturm, Amir A. Khan, John D. Mellinger, Sabha Ganai

Introduction: Current practice guidelines for pancreas cancer recommend resection at high-volume centers after multidisciplinary discussion. However, while centralization of care is promoted, there is evidence that surgery is not being recommended for a majority of patients at a national level. We hypothesized that there may be regional and rural-urban differences in resection for early-stage pancreas cancer. **Methods:** National Cancer Database pancreas cancer cases with cT1 or cT2 staging diagnosed between 1998 and 2012 with a valid rural-urban designation (n=28,451) were analyzed. The proportions of patients who had surgical resection by rural vs. urban status and region were compared. The Chi-square test for independence was used to compare the proportion of patients who had surgical resection by rural vs urban status and region, and choropleth maps were developed to display tertiles of resection by region. Cox proportional hazards analysis was used to determine rural-urban differences in survival after resection accounting for demographic, geographic, clinical, and facility characteristics. **Results:** Characteristics of early stage pancreas cancer patients varied by demographic, geographic, and facility characteristics. Among cT1-T2 cancers nationwide between 1998 and 2012 (n=28,451), there were 12,486 resections (43.9%). While median distance to surgical management was different (rural 50.2 miles vs. urban 8.9 miles), resection rates were similar between rural and urban county of residence (43.8% vs. 43.9% nationally; p=0.84), except in the Middle Atlantic region where surgical resections were higher among rural patients compared to urban patients (55.3% vs. 49.1%; p=0.02). Similar median time to surgery was noted for rural and urban residence. Figure 1 demonstrates regional differences in resection rates. In adjusted analysis, there was a modest, but statistically significant increased risk for death among rural patients who underwent resection in comparison to urban patients (HR 1.11; 95% CI, 1.04-1.19, p<0.0001). **Conclusions:** While cT1 and cT2 pancreatic cancers are potentially resectable, less than half of these early-stage cancers underwent surgery. There were no important differences in resection rates between rural and urban county of residence. This suggests that access disparities may not completely explain the low rates for pancreatotomy which may be a result of failure of surgical referral and ability to resect in both locales. There may also be selection bias present from differential access to Commission on Cancer accredited facilities with under-reporting of rural cases that did not undergo resection. Differences in survival after resection are present among rural patients. Further analysis of barriers to referral for pancreatic resection for rural and urban patients is warranted.